PRE-IDEATHON TEMPLATE

GENESIS IDEATHON

TEAM & DOMAIN

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- Selected Domain – #6: Healthcare, Biotechnology and Human well-being
- Problem Statement – AI-Based Predictive and Adaptive Ventilator Support System for Critical Care Units.



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DOMAIN PROBLEM IN OUR WORDS

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What exactly needs to be solved?

- Mechanically ventilated patients can experience sudden changes in respiratory condition.
- As the patient’s condition changes, ventilator settings may require timely manual adjustment by clinicians.
- Clinicians may be managing multiple critical patients simultaneously, making continuous individual monitoring difficult.
- A patient's deterioration may begin between routine clinical assessments, creating a need for timely response.

Why is it important right now

- Delayed recognition of respiratory deterioration can increase the risk of complications in critically ill patients.
- Continuous manual monitoring and adjustment adds to the workload of critical-care clinicians.
- The need for timely respiratory support is especially important in high-dependency and resource-constrained critical-care environments.
- Existing advanced automated ventilation systems are not universally available across all healthcare settings.

Bottom line:

We need a way to recognize changing respiratory conditions and provide timely ventilation support when immediate clinician attention is unavailable, while keeping clinicians in control.

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WHO IS AFFECTED?

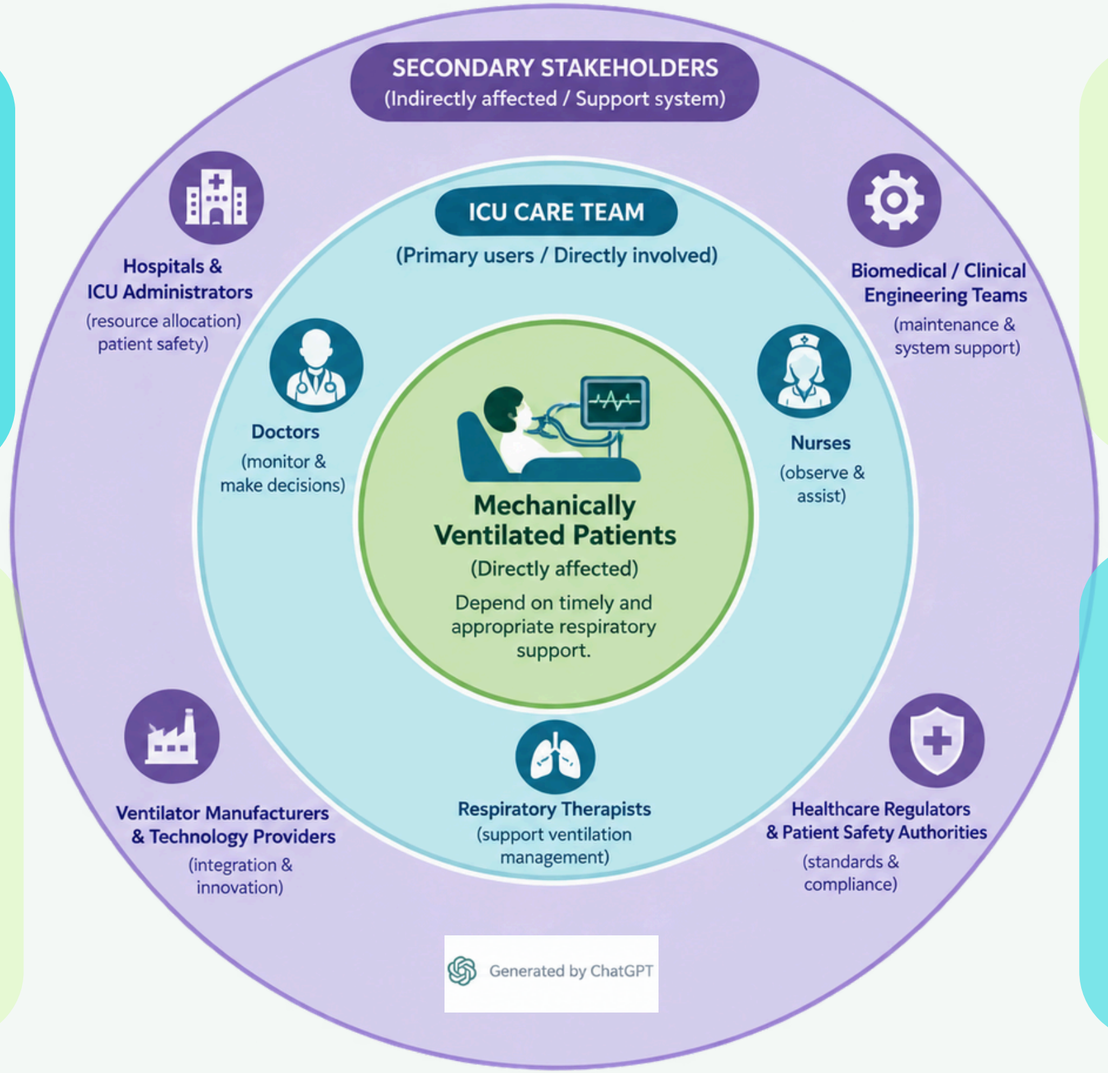
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PATIENT IMPACT

- **PRIMARY AFFECTED:** Mechanically ventilated ICU patients.
- Respiratory conditions can change rapidly,requiring timely adjustment of ventilatory support

CLINICAN IMPACT

- **DIRECTLY AFFECTED**: ICU doctors, nurses, and respiratory therapists.
- Managing multiple critically ill patients can make continuous monitoring and timely adjustments challenging.



CARE IMPACT

- **AFFECTED BY DELAYED RESPONSE**: Patients requiring continuous respiratory support.
- Early deterioration may occur between routine clinical assessments.

HEALTHCARE SYSTEM IMPACT

- **AFFECTED SETTINGS**: ICUs and hospitals, especially resource-constrained facilities.
- Limited access to advanced automated ventilation can create gaps in timely respiratory support.

CURRENT SITUATION & CONTEXT

Existing Scenario and Background

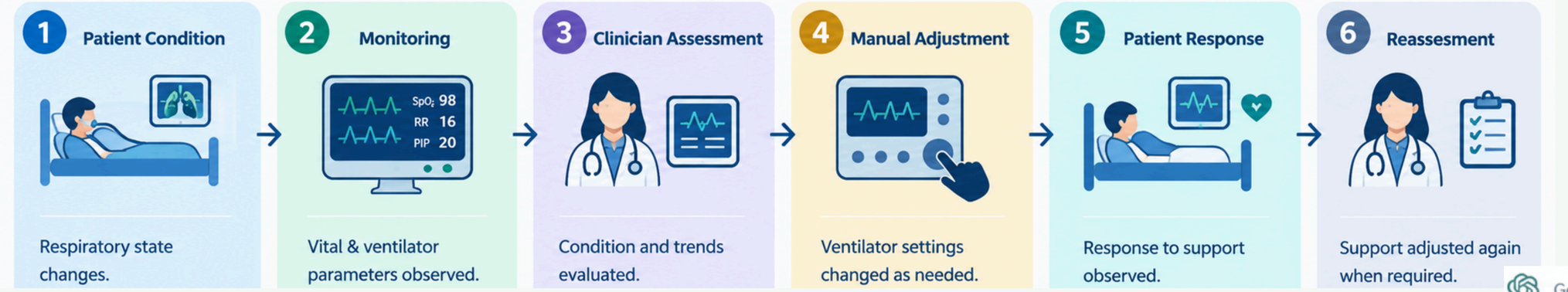
- Mechanical ventilation supports critically ill patients who cannot breathe adequately on their own.
- Ventilators continuously monitor respiratory parameters and provide controlled breathing support.
- Clinicians monitor the patient's condition and adjust ventilator settings as needed.
- Some advanced ventilators offer automated support, but capabilities vary across devices and healthcare settings.

Relevant Data/Facts

- Respiratory conditions in critically ill patients can change rapidly, requiring frequent reassessment.
- Ventilator settings directly influence oxygenation, ventilation, and patient safety.
- Closed-loop ventilation has demonstrated the feasibility of automated ventilator adjustment.
- An opportunity remains to use continuous patient data with predictive detection of deterioration early.

Current Workflow

How ventilation is managed today



Generated by ChatGPT

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KEY CHALLENGES

Main difficulties, bottlenecks, root causes and consequences

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Main Difficulties

- Rapid changes in patient respiratory condition
- Need for timely ventilator adjustment
- Limited clinician attention across multiple ICU patients
- Difficulty recognizing early signs of deterioration
- Continuous monitoring and adjustment can be challenging

Root Causes

- Patient condition can change continuously and unpredictably
- Ventilator adjustments depend on timely clinical assessment
- Clinicians cannot continuously focus on individual patients
- Subtle deterioration may develop between clinical assessments
- Patient data may change faster than routine clinical assessments can capture

Consequences if Left Unsolved

- Delayed recognition and intervention
- Inadequate or delayed respiratory support
- Increased workload and risk of missed changes
- Greater risk of severe or life-threatening deterioration
- Reduced opportunity for early preventive intervention

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EXISTING SOLUTIONS / APPROACHES

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1. Existing Methods / Systems / Alternatives

- **Manual ventilator management** — Clinicians monitor the patient and adjust ventilator settings as needed.
- **ICU monitoring & alarm systems** — Continuously monitor physiological and ventilator parameters and alert clinicians to abnormal conditions.
- **Closed-loop ventilation systems** — Some advanced ventilators automatically adjust selected settings based on measured patient/ventilator data.
- **Clinical protocols & decision-support tools** — Provide predefined targets, guidelines, or recommendations for ventilator management.

2.Limitations

- Manual management
- Monitoring & alarms:
- Closed-loop systems
- Protocols & decision support

3.Gap / Open Challenge.

A gap remains for an accessible system that continuously analyzes existing patient and ventilator data, predicts impending respiratory deterioration, and provides timely, safety-constrained automatic support while keeping the clinician in control.”

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PROBLEM INSIGHTS & OPPORTUNITY AREAS

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1.Hardware Requirements

- **Existing ICU Ventilators & Patient Monitors:** Provide patient vitals and ventilator readings for analysis.
- **GPU-Enabled Server or Cloud Computing**: Processes patient data and runs the predictive AI model.
- **ICU Computers/Workstations**: Allow clinicians to view predictions, risk alerts, and recommended adjustments.

2.Software Requirements

- **AI/ML – Python + PyTorch:** Develop a predictive model to identify early signs of respiratory deterioration.
- **Backend – FastAPI:** Processes incoming data and connects the AI model to the user interface.
- **Database – PostgreSQL:** Stores historical patient measurements, predictions, and system logs.
- **Dashboard – React:** Displays patient status, predicted risks, and alerts for clinicians.
- **System Integration:** Connects to compatible ventilator data interfaces and hospital systems where available.

3.Other Resources

- **Clinical Experts:** Intensivists and respiratory therapists to guide clinical requirements and safety limits.
- **Technical Team:** AI/ML engineers and software developers to build and test the system.
- **Datasets:** MIMIC-IV and simulated patient data for model development and testing.
- **Safety & Validation**: Clinical evaluation, cybersecurity testing, and applicable medical-device regulatory compliance.

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INITIAL DIRECTION & EXPECTED OUTCOME

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1.What our team intends to do?

- Develop a predictive AI model to identify early signs of respiratory deterioration
- Analyze existing patient and ventilator data continuously.
- Enable automatic ventilator adjustments within predefined safety limits.

What We Want to Improve

- Early recognition of potential respiratory deterioration.
- Timeliness of ventilator adjustments.
- Reduce the burden of continuous monitoring on clinicians.

Expected Success

A predictive AI-powered ventilation support system that detects early signs of respiratory deterioration, enables timely and safe ventilator adjustments, reduces clinicians' workload, and delivers more responsive, personalized support while keeping clinicians in control.